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Blockchain Ledgers for Tracking Datasets Used in Training AI Models

ABSTRACT

Owners and providers of artificial intelligence (AI) or machine learning (ML) models have an interest in tracking the datasets used in the training of their models for compliance and/or troubleshooting and in ensuring that high quality datasets are used for training the model. Dataset creators or owners have an interest in knowing if their dataset is being used to train an AI model. This disclosure describes blockchain techniques that capture and maintain the lineage and integrity (tamper-evidence) of a training dataset used for specific versions of AI/ML/generative-AI models. Given a training dataset, its hash is containerized along with a link to its location and its usage policy, and the container is uploaded to a blockchain ledger. The transparent, append-only blockchain record enables model owners to track their training datasets and to comply with usage policy, and enables content owners to detect content usage. The evolution of an AI model can be tracked against training datasets, enabling dataset optimization/de-biasing and reproducible AI-model behavior.

KEYWORDS

- Artificial intelligence (AI)
- Generative AI (GenAI)
- Large language model (LLM)
- Data lineage
- Training dataset
- Blockchain ledger
- Immutable ledger
- Append-only ledger
- Tamper evident supply chain
- Software supply chain
- Secure supply chain

BACKGROUND

Large datasets are used to train artificial intelligence (AI) models such as large language models (LLMs). It is important that the training datasets be accurate, up to date, and diverse, as the performance of the AI models can depend on the quality of training datasets. Generally, the creator or owner of any dataset has an interest in knowing if their dataset is being used to train an AI model. The lineage of a dataset used for AI training is thus of interest to both the creator/owner of the dataset and the owner, operator, developer, or creator of the AI model. It is also important to ensure that training data is representative of the real world and does not contain biases or inaccuracies.

Append-only blockchain ledgers enforce time-ordered appending of data. Once data is appended to the blockchain, it is infeasible to edit such data and therefore, appended data can be considered as immutable.

DESCRIPTION

This disclosure describes blockchain techniques that enable the capture and robust maintenance of the lineage of a dataset being used to train AI models. Original sources and location(s) of the datasets, usage rights/policies, and modifications to the dataset are tracked using a blockchain.

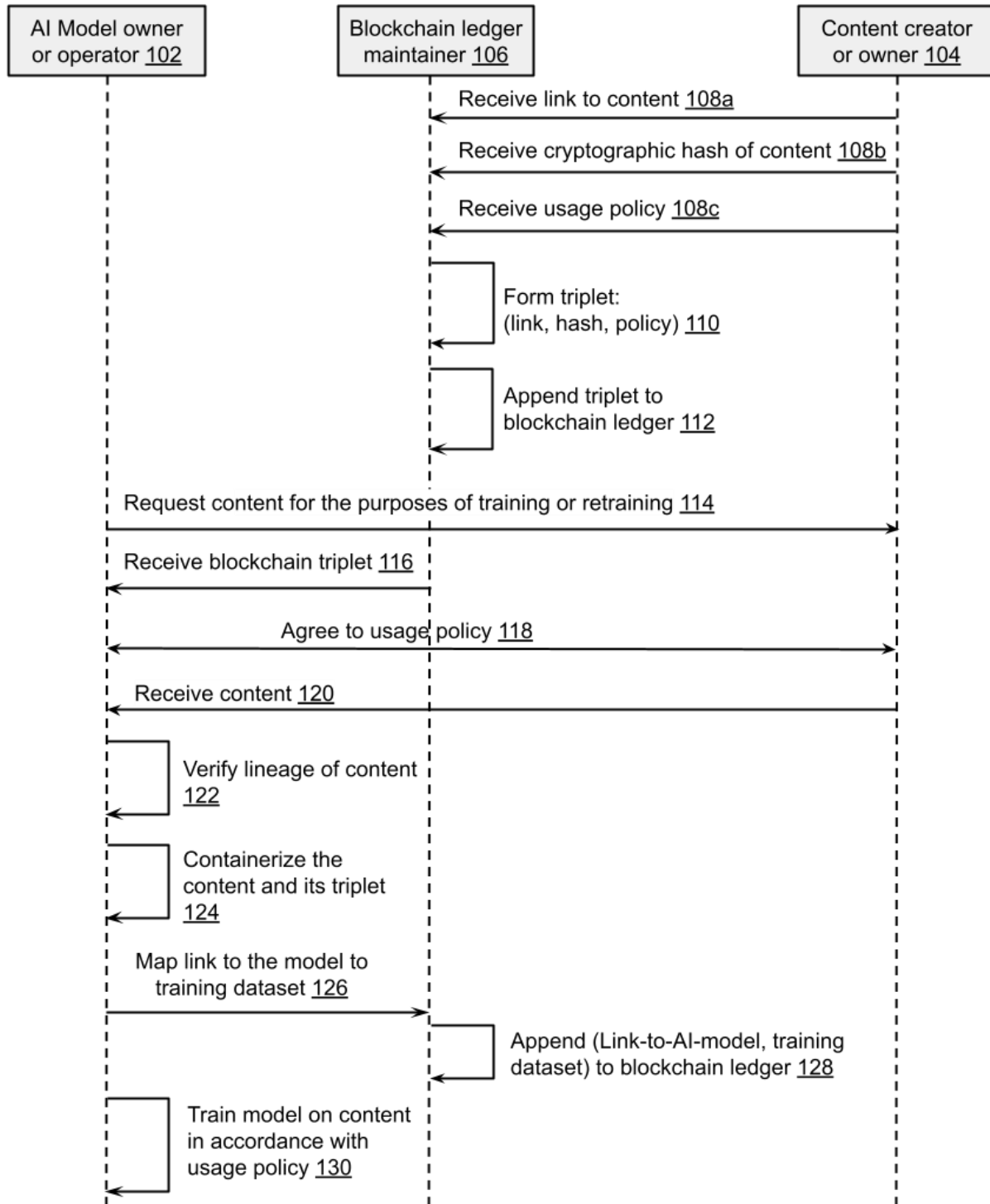


Fig. 1: Blockchain ledgers for tracking datasets used in training AI models

Fig. 1 illustrates blockchain ledgers for tracking datasets used in training AI models. An AI model owner or operator (102) seeks high quality content to train their model. Such content is available with a content creator or owner (104). The AI model owner/operator seeks assurance that the content being used as training dataset is genuinely from the content creator/owner. The content creator/owner seeks assurance that the content is used to train the AI model in accordance with a usage policy.

The content creator/owner uploads to a blockchain ledger maintainer (106) a link to particular content (108a), a cryptographic hash of the content (108b), and a usage policy (108c). The blockchain ledger maintainer forms a triplet (110) of {link, hash, policy} and cryptographically appends the triplet to an append-only blockchain ledger (112). The use of an append-only blockchain ledger ensures that the training dataset lineage can be transparently verified.

The blockchain ledger can be private or public. As explained below, in either case it can be used for model reproducibility, tuning/troubleshooting, tracking royalty payments for training data, compliance reporting, and other purposes. The AI model owner/operator requests the particular content from the content owner/operator for the purposes of training or retraining their model (114). The AI model owner/operator receives the blockchain triplet associated with the particular content (116). The AI model owner/operator and the content creator/owner agree to the usage policy (118). The AI model owner/operator receives the content from the content creator/owner (120).

Applying cryptographic techniques to the blockchain triplet, the AI model owner/operator verifies the lineage (provenance) of the content (122), based on the identity of triplet as established by the content creator/owner. The received content and the triplet can be

containerized (124), e.g., can be made to travel together for the purposes of tracking the evolution of the AI model. The AI model owner/operator maps a link (or reference) to the AI model to the training dataset and sends the mapping to the blockchain ledger maintainer (126). The blockchain ledger maintainer appends the mapped pair <link-to-AI-model, training dataset> to the ledger (128). The AI model owner/operator trains their model on the received content in accordance with the usage policy embedded within the blockchain triplet (130).

An AI model that draws upon multiple content creators/owners can follow a similar procedure for each distinct content creator/owner. The procedures of triplet {link, hash, policy} formation and containerization can be performed by the content creator/owner, the blockchain ledger maintainer, or the AI model owner/operator.

Some advantages of the described techniques include:

- **Framework:** The techniques can serve as a trusted framework for interaction between content creators/owners and AI model owners/operators.
- **Tracking:** Evolving versions of an AI model can be tracked against their corresponding training datasets as follows. Pointers (links) to the set of training datasets used to train a given model can be captured and packaged chronologically alongside the model. Issues of performance degradation, improvement, or bias can be traced to particular training datasets.
- **Reproducibility:** The techniques provide the capability of model checkpointing, model versioning against time, and model restoration. Rollback to a version or state of the model that performed particularly well is made possible. Instead of retraining a model from scratch, a model owner/operator can train the model anew from a rolled-back point. In this manner, the model owner's investment in training the model is protected.

- **Re-training:** A change in a training dataset reveals itself in the blockchain ledger as a change in its hash. An AI model owner/operator can elect to retrain their model upon detecting a change in a dataset that was previously used to train the model.
- **Transparency:** The described append-only ledger-based approach to tracking training datasets provides security, integrity, and transparency. A third-party can examine the ledger.
- **Integrity and tamper-evidence:** The described append-only ledger based mechanism can ensure that the integrity of the training datasets integrity is preserved and remains untampered. It can ensure that the datasets are not poisoned or maliciously modified.

The evolution of an AI model can be tracked against its training datasets, enabling optimization of training datasets, reproducible AI-model behavior, re-trainable AI models, etc.

CONCLUSION

This disclosure describes blockchain techniques that capture and maintain the lineage and integrity (tamper-evidence) of a training dataset used for specific versions of AI/ML/generative-AI models. Given a training dataset, its hash is containerized along with a link to its location and its usage policy, and the container is uploaded to a blockchain ledger. The transparent, append-only blockchain record enables model owners to track their training datasets and to comply with usage policy, and enables content owners to detect content usage. The evolution of an AI model can be tracked against training datasets, enabling dataset optimization/de-biasing and reproducible AI-model behavior.

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